DATA ITEM DESCRIPTION	
2. Title SHIPS, BOOKLET OF INCLINING EXPERIMENT DATA	1. ID No. DI-MISC-80904

3. Description/Purpose

3.1 The BOOKLET OF INCLINING EXPERIMENT DATA consists of two parts. The Inclining Experiment Report (Part 1) contains the observations and calculations leading to the determination of the displacement and location of the (Continued on Page 2)

4.Approv Date	5. OPR	6a.DTIC	6b.GIDEP
891003	SH/SEA 55W4		

7. Application/Interrelationship

7.1 This DID contains the format and content preparation instructions for the data product generated by the specific and discrete task requirements as delineated in the contract.

7.2 This DID supersedes UDI-E-23202 and UDI-E-23255.
(Continued on Page 2)

8. Approval Limitation	9a. Application Forms See page 2	9b.AMSC No. N4837

10. Preparation Instructions

10.1 Reference Documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions shall be as specified in the contract.

10.2 Format. The report shall be in the format specified in 096-2.129 of NAVSEA S9086-C6-STM-000/CH-096.

10.3 Content. The report shall contain the following as appropriate:

10.3.1 Part 1 - Inclining Experiment Report. The inclining experiment report shall contain the following pages as applicable:
(Continued on Page 2)

11. Distribution Statement

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Block 3, Description/Purpose (Continued)

center of gravity of the ship in light condition. Stability Data (Part 2) for surface ships and Stability and Equilibrium Data for Submarines contain data relative to the characteristics of the ship in the operating conditions. Preliminary reports of Parts 1 and 2 permit evaluation of the ship's stability and reserve buoyancy and may indicate the necessity for action to improve the ship, a change in policy or weight control or additional inclining experiments.

Block 7, Application/Interrelationship (Continued)

7.3 The reference document cited herein may be obtained by writing to:

Naval Publications and Forms Center
5801 Tabor Ave.
Philadelphia, PA 19120 Attn: Customer Service

Block 9a, Applicable Forms (Continued)

9.a Standard NAVSEA forms used in the preparation of this report may be obtained by contacting:

Naval Sea Systems Command
Code 55W4 Stability Division
NC-3 Rm 3W40
2531 Jefferson Davis Highway
Arlington, VA 22202

Block 10, Preparation Instructions (Continued)

Form Number NAVSEA 9290/6-	Form Title
1	Title Sheet - Inclining Experiment Report, (Part 1), For Surface Ships and Submarines
2	Armament, Boats, Submarine Batteries, Ballast
3	Ship in Condition A - Light Ship
4	Change in Condition A Weight Since Inclining
5/6	Displacement and Center of Gravity As Inclined
7	Functions of Wedge Areas (as required)
8	Displacement and Center of Gravity in Conditions A and A-1
9	Weight Movements and Inclinations
10	Plot of Tangents

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Block 10, Preparation Instructions (Continued)

Form Number NAVSEA 9290/6-	Form Title
11	Weights to Complete
12	Weights to Deduct
13	Weights to Relocate
14	Vertical Moment of Free Surface "As Inclined"
15	Diagram Showing Location of Draft Marks
16	Remarks and Miscellaneous Calculations

10.3.2 Part 2 - Stability Data for Surface Ships Only. The stability report for surface ships shall contain the following pages as applicable:

Form Number NAVSEA 9290/6-	Form Title
17	Title Sheet - Inclining Experiment Report, (Part 2), For Surface Ships Only
2	Armament, Boats, Submarine Batteries, Ballast
3	Ship in Condition A - Light Ship
4	Change in Condition A Weight Since Inclining
	*Loading Conditions:
18	Ship in Condition ___ (Small Free Surface Effect), or
19	Ship in Condition ___ (Large Free Surface Effect) Excessive Trim in Loading Condition
20	Displacement and Other Curves
21	Cross Curves of Stability
15	Diagram Showing Location of Draft Marks
22	Approximate Change in Metacentric Height Due to Added Weight
23	Summary of Load Items in Condition ___
24	Details of Load Items in Condition ___
25	Correction to Righting Arms for Free Surface Effect in Condition ___
26	Tank Capacities
27	Compartment Capacities
28	Table of Frame Spacing
16	Remarks and Miscellaneous Calculations

* The standard loading conditions are:

Condition	Description
A	Light Ship
A-1	Light Ship (w/o permanent ballast)
B	Minimum Operating
C	Optimum Battle
D	Full Load Departure

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Block 10, Preparation Instructions (Continued)

10.3.3 Part 2 - Stability and Equilibrium Data for Submarines. The stability and buoyancy report for submarines shall contain the following pages as applicable:

Form Number NAVSEA 9290/6-	Form Title
29	Title Sheet - Inclining Experiment Report, (Part 2), Stability and Equilibrium Data for Submarines
30	Load to Submerge Determination
2	Armament, Boats, Submarine Batteries, Ballast
3	Ship in Condition A - Light Ship
31	Detail Changes in Condition A and Submerged Displacement Since Last Inclining and Trim Dive
32	Load to Submerge at Time of Trim Dive
33	Details of Load on Trim Dive
34	Ship in Condition ___ - Surface Diving Trim
35	Ship in Condition ___ - Submerged
36	Variable Ballast in Condition ___
20	Displacement and Other Curves
21	Cross Curves of Stability
15	Diagram Showing Location of Draft Marks
37	Variable Load in Conditions N and M
38	Details of Variable Load in Condition N and M
39	Water Ballast in Main Ballast, Fuel Ballast and Safety Tanks
40	Residual Water, Water Seal, and MBT Lead Corrections
41	Equilibrium Polygon
42	Points for Equilibrium Polygon
	*Equilibrium Conditions
43	Light No.1 & Light No.2
44	Heavy No.1 & Heavy No.2
45	Heavy No.1 (With Mines) & Heavy No.2 (With Mines)
46	Heavy FWD No.1
47	Heavy FWD No.2
48	Heavy Aft
49	Details of Load for Equilibrium Conditions
50	Plot of Minimum GM While Trimming Down
51	Conditions While Trimming Down
52	Ship in Condition ___ Surface Diving Trim, Ballast Tanks Flooded, One Side Only
28	Table of Frame Spacing
--	Plot of Hull and Material Compression vs Percent of Test Depth
16	Remarks and Miscellaneous Calculation

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Block 10, Preparation Instructions (Continued)

10.3.4 Engineering Judgement. An opinion by the inclining activity of the reliability of the experiment.

10.3.5 Graphics. Displacement and other curves drawing.

10.3.6 Photographs. Photographs of the vessel topside installations, particularly the location of inclining weights, masts, shipyard gear and draft marks, forward/aft and port/starboard, at zero inclining moment.

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